Md Akib Haider

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Research Interests: Human-Centric AI, Computer Vision, Machine Learning

EDUCATION

- **Islamic University of Technology (IUT)** Aug 2022 – Present
  B.Sc. in Computer Science and Engineering
  – **CGPA**: 3.91/4.00
- **Notre Dame College (NDC)** 2019–21
  Higher Secondary School Certificate (HSC), Science
  – **GPA**: 5.0/5.0 **(Government Merit Scholarship)**.

RESEARCH EXPERIENCE

- **DPPC: Discriminative Pairwise Predictive Correction with Comparative Prompts for Training-Free Few-Shot Class-Incremental Learning** Under Review, WACV
  – We introduced Tri-Level Calibration (TRIC) in FSCIL, semantic calibration within the language branch, fuses CLIP visual evidence on a common score axis, and inter-modal calibration between resulting visual and semantic evidence for fully inference based classification.
- **Prototype-Based Test-Time Prediction Calibration for Few-Shot Object Detection** Under Review, WACV
  – We proposed a Patch-Token Prototype Calibration method by encoding query images via a frozen foundation-model encoder for fully inference-based confidence recalibration of any CNN-based detector.
- **Diagnosing Prototype Estimation in Frozen-Feature Few-Shot Class-Incremental Learning: Training-Free Geometry Selection and Unlabeled-Query Prototype Rectification** Under Review, WACV
  – We proposed finite-shot prototype estimation as the dominant recoverable error source in frozen few shot class-incremental learning, and complements it with a transductive correction for unlabeled query settings.
- **Probabilistic Feature Imputation and Uncertainty-Aware Multimodal Federated Aggregation**
  Published in MIDL | `Paper` 2026
  – We designed a probabilistic federated learning framework that replaces point estimates with calibrated distributions for uncertainty-aware local fusion and weighted aggregation across missing modalities.
- **LoRA-MME: Multi-Model Ensemble for Code Comment Classification**
  Published in ICSE-NLBSE | `Paper` 2026
  – We proposed a weighted stacking ensemble learning method while fine-tuning four complementary code encoders to solve multi-label classification across different programming languages in total 18 classes; achieving better classification performance over baseline.

WORK EXPERIENCE

- **Undergraduate Research Intern** Jun 2026 – Present
  University of Illinois Urbana Champaign, SRSE program, `Sharifa Sultana's Lab`
- **App Design and Development (Remote)** Feb 2025 – Apr 2025
  SwadeshBazar, Chelmsford, UK
  – Built Android and web version in Flutter and Next.js for a grocery delivery and warehousing start-up featuring product browsing, cart, dashboard, designed with Rave and payment gateway for secure transactions.
- **Industrial Training on Mobile Application Development** Sep 2025 – Oct 2025
  Battery Low Interactive Ltd, Dhaka, Bangladesh
  – Built an Anime streaming and sharing app using Flutter and Dart, featuring state management, API integration, UI/UX design, testing, and deployment | **code**.

PROJECTS

- **29 Game State Detection and Move Analysis using Video-Scene Understanding** 2026
  Python, PyTorch, OpenCV, OpenVINO, YOLOv26 | code
  - Built a system to detect game states, analyze gameplay, and predict moves in the card game 29 under low-resource settings.
- **Better_SIS, University Management System** | *Dart, C++, Swift, Flutter* | code 2025
  - Developed Student Information System (SIS) of IUT, featuring academic activities and internal service management for students, teachers, and staff.
- **Student Driven Market Place** | *Flutter, Gemini 2.5, Open Street Map* | code 2025
  - Created a virtual institutional marketplace where students can buy and sell their products within the community, donate items to those in need, and easily manage product deliveries.
- **Animeverse** | *Flutter, Lmstudio, Dart, Docker, Jikan Api, Rive* | code 2025
  - Built an anime streaming platform featuring AI agents, personalized recommendations, anime tracking, discussion forums, downloads, local sharing, and social networking, creating a virtual anime club.
- **Xchange, File Sharing System** | *Socket Programming, Java* | code 2024
  - Developed a secure and efficient peer-to-peer file-sharing system enabling fast file transfers over a local network without requiring internet access.
- **E-Commerce** | *Dart, C++, Rive, Flutter, Docker* | code 2024

GRANTS AND FELLOWSHIPS

- **ACM SIGHPC Travel Grant**
  - HPC Immersion Program, Attended SC 2025 in St. Louis, Missouri, USA ($3000). 2025
  - Lightning Talk on "Data, Model, and Tensor Parallelism for efficient MRI Translation".

ACHIEVEMENTS

Contests & Awards

- **NLBSE Tool Competition** winner, co-located workshop with ICSE-26. 2026
- **Programming Contest**
  - **26th** (among 110 teams), IUT 12th National ICT Fest Programming Contest. 2026
  - **80th** (among 162 teams), Uttara University Inter-University Programming Contest. 2025
  - **6th**, Intra-IUT Junior Programming Contest. 2024
  - **52nd** (among 115 teams), IUT 11th National ICT Fest Programming Contest. 2024
  - **102nd** (among 317 teams), ICPC Asia Dhaka Regional Onsite Contest. 2024
  - Personal best: Country **rank 37**, Global **rank 1536** in Codeforces | profile. 2022-26
- **Hackathon**
  - **Champion** in Coderush 2.0, IUT-ICT Fest Hackathon | code 2025
- **Software Sub-Team Member, Project Altair** at IUT Robotics Society. 2023–24
  - **Global 6th place** and **Best Science Team award** in International Rover Challenge (IRC).

Games & Sports

- **Captain,** Inter University Chess Team of IUT | *FIDE Blitz Rating: 1553*. 2025–26
- All Bangladesh **Champion,** Intra-IUT Annual Drought and Scrabble Competition. 2026
- All Bangladesh **2nd Runner Up,** Intra-IUT Annual Table Tennis Competition. 2026
- All Bangladesh **rank 49** at 37th National Sub-Junior Chess Championship. 2018

Literature & Creative Work

- **Delegate (Writing)** at 6th International Children Film Festival. 2013
- **UNICEF-Meena Media Award** winner for creative writing in print media. 2012–13

LEADERSHIP & EXTRA CURRICULAR ACTIVITIES

- **Instructor, IUT ACM Community** Aug 2024 – Present
  - Mentor for sophomore Competitive Programming aspirants.
  - Organized and conducted weekly classes.
- **Organizer, Intra-IUT Junior Programming Contest** 2025
  - Event head at *Coderush 2.0*, organized by IUT Computer Society.
- **Vice President, Notre Dame English Club** 2019–22
  - Organized *6th National English Carnival*, Literature & Pop-Culture Quiz: *Litbuzz 3.0*.
- **Problem Setting**
  - Intra-IUT Junior Programming Contest | problemset 2025
  - Abid Hasan Intra University Programming Contest | problemset 2025

Sports: Passionate about playing table tennis and chess.

REFERENCES

- **Dr. Md. Hasanul Kabir**
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